Industry Stock Market Reactions Around the Russia–Ukraine War

Final Report

STATS 405
Group 6
Sam G., Nathan R., Shaina W., Yue W., Jennie Z.

Abstract. This project studies how selected industry and regional exchange-traded funds reacted around Russia's invasion of Ukraine on February 24, 2022. We use SPY as the broad U.S. market benchmark and compare Energy (XLE), Aerospace and Defense (ITA), Airlines (JETS), Financials (XLF), Technology (XLK), Materials (XLB), Europe (VGK), and Germany (EWG). The analysis combines an event-study model with a difference-in-differences model. The event study shows that Energy and Aerospace and Defense produced the strongest market-model cumulative abnormal returns, while Germany, European equities, Financials, and Technology underperformed. The DiD model gives a weaker statistical result: the treatment interaction is positive but not statistically significant. Together, the evidence suggests that the war-related reaction was sector-specific and front-loaded, but the results should be interpreted as abnormal market patterns rather than strict causal proof.

1 Introduction

Financial markets often change rapidly during global conflict because investors reassess supply chains, commodity prices, inflation, military spending, interest-rate expectations, and future growth. One important example is Russia's invasion of Ukraine on February 24, 2022. The event created uncertainty around energy supply, European growth, sanctions, transportation routes, and defense spending. Although broad equity markets declined around the event, the effect was not evenly distributed across industries.

This project asks a specific empirical question: which industries and regions outperformed or underperformed relative to the broader U.S. stock market around the Russia–Ukraine War escalation? Rather than predicting future prices, the goal is to measure whether the observed sector movements were larger or smaller than what would be expected from normal market behavior.

The central intuition is straightforward. If every sector falls because the whole market falls, that does not necessarily mean the war affected each sector differently. To isolate sector-specific behavior, each ETF is compared against SPY. This lets the analysis focus on abnormal performance: the part of a sector's return that cannot be explained by the broad market alone.

The project is also motivated by the difference between economic intuition and statistical evidence. It is easy to say that energy and defense should benefit from a war shock, but the report tests whether those sectors actually behaved differently after accounting for the broader market. This is why the analysis combines visual evidence, cumulative abnormal returns, and regression output instead of relying only on raw price movements.

A second motivation is regional exposure. The war took place in Europe, and European markets faced direct risks from energy dependence, sanctions, trade disruption, and uncertainty about growth. Including VGK and EWG allows the analysis to compare industry exposure with geographic exposure. Germany is especially relevant because its economy was closely connected to Russian energy supply before the invasion.

2 Data and Methods

2.1 Data Collection and ETF Selection

Daily price data were downloaded from Yahoo Finance for the period from August 24, 2021 to August 24, 2022. The data collection script queries Yahoo Finance's chart endpoint for each ticker, converts the response into a pandas DataFrame, removes rows with missing close prices, and saves both individual ticker files and combined price files. The final dataset contains daily close prices for SPY and the eight sector or regional ETFs.

The project code is organized into five files. The first script downloads and saves the ETF price data. The second script estimates the event-study model and writes the CAR tables. The third script creates the event-study figures. The fourth script estimates the DiD regression. The fifth script creates the DiD plots. This workflow makes the analysis reproducible because each output table or figure is produced from a specific code file rather than manually assembled.

Table 1: ETF Selection and Project Role

Ticker	**Role**	**Reason Included**
SPY	Benchmark	Broad U.S. equity market comparison
XLE	Treatment sector	Energy supply and oil-price exposure
ITA	Treatment sector	Defense spending and aerospace exposure
JETS	Control sector	Travel, fuel cost, and airspace disruption exposure
XLF	Control sector	Financial-market and sanctions sensitivity
XLK	Control sector	High-beta growth and risk-off exposure
XLB	Control sector	Materials and commodity-linked production
VGK	Regional ETF	Broad European market exposure
EWG	Regional ETF	Germany-specific European exposure

The event date is February 24, 2022. The pre-event estimation period runs from August 24, 2021 to January 24, 2022, which is used to estimate normal relationships between each ETF and SPY before the invasion. This estimation window is separated from the event date so that the market model is fitted on pre-event behavior rather than on returns already affected by the invasion. The event-study windows are:

1. Short window: $[-10, +10]$ trading days around the event.
2. Main window: $[-30, +30]$ trading days around the event.
3. Post-event window: $[0, +60]$ trading days after the event.

2.2 Event Study Model

The event study is the primary model. It estimates how each ETF would normally move with the market, then compares the observed return to that expected return. Daily log returns are calculated from ETF close prices. For each non-SPY ticker, the market model is estimated as:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t}, \tag{1}$$

where $R_{i,t}$ is the daily return of ETF i, $R_{m,t}$ is the daily return of SPY, α_i is the ETF's intercept, and β_i measures how sensitive the ETF is to broad market movements. The abnormal return is:

$$AR_{i,t} = R_{i,t} - (\alpha_i + \beta_i R_{m,t}). \tag{2}$$

Cumulative abnormal return (CAR) adds abnormal returns across a chosen event window:

$$CAR_i = \sum_t AR_{i,t}. \tag{3}$$

CAR is the most important outcome in this report because it measures whether an ETF performed better or worse than expected after controlling for broad market movement. A positive CAR means the ETF outperformed its expected return during the event window; a negative CAR means it underperformed. This does not automatically prove causality, but it identifies where market behavior was unusually strong or weak relative to the benchmark model.

The event study has two advantages for this project. First, it is designed for a clearly dated shock, and February 24, 2022 is a natural event date. Second, it can capture fast market reactions that happen within days or weeks. Its main limitation is that other macroeconomic forces can still overlap with the event window, especially inflation and energy prices in early 2022.

2.3 Difference-in-Differences Model

The second model is a difference-in-differences (DiD) regression. This model compares a treatment group against a control group before and after the event. The treatment group contains sectors expected to benefit most directly from the war shock: Energy (XLE) and Aerospace and Defense (ITA). The control group contains JETS, XLF, XLK, XLB, VGK, and EWG.

The DiD model estimates:

$$Y_{i,t} = \gamma_0 + \gamma_1 Post_t + \gamma_2 Treatment_i + \gamma_3 (Post_t \times Treatment_i) + u_{i,t}. \tag{4}$$

The key coefficient is γ_3, the interaction term. If it is positive and statistically meaningful, the treatment group improved more after the event than the control group. In this project, DiD is best understood as a robustness check rather than the main model because the war shock was fast-moving and sector-specific.

The DiD approach relies on a parallel-trends idea: before the event, the treatment and control groups should not be moving in completely different ways. If the groups were already diverging before the invasion, the post-event comparison would be harder to interpret. Because our treatment group contains only two ETFs and the event window is short, the DiD result should be read cautiously.

3 Results

3.1 Reading the Output

Before interpreting individual sectors, it is useful to separate three return measures. Cumulative return is the raw total return of an ETF during the window. Simple CAR subtracts SPY's cumulative return from the ETF's cumulative return. Market-model CAR goes one step further by adjusting for the ETF's normal beta with SPY. Because different sectors have different market sensitivity, the market-model CAR is the strongest metric in this report.

3.2 Market-Model Estimates

Table 2 reports the estimated market-model parameters. Beta measures how strongly each ETF typically moves with SPY. XLK has the highest beta at 1.320, meaning Technology was the most sensitive to broad market swings in the estimation period. EWG and VGK both have betas around 0.751, meaning they moved less than the broad U.S. market on average. XLE has the largest positive alpha, which suggests Energy was already trending upward before the invasion.

Table 2: Market-Model Estimates

Ticker	**Description**	**Alpha**	**Beta**
EWG	German equities	-0.000785	0.751
ITA	Aerospace and Defense	-0.000162	0.931
JETS	Airlines	-0.000809	1.128
VGK	European equities	-0.000399	0.751
XLB	Materials	+0.000078	0.852
XLE	Energy	+0.002845	1.051
XLF	Financials	+0.000216	0.989
XLK	Technology	+0.000032	1.320

3.3 Main Event Window

The main $[-30, +30]$ window gives the clearest comparison across sectors. SPY returned -3.4% during this window, showing broad market weakness. XLE returned +24.5%, producing a simple CAR of +27.9% and a market-model CAR of +10.7%. ITA returned +5.5%, producing a market-model CAR of +9.6%. These two sectors were the strongest relative performers.

The weakest performer was EWG, which returned -17.3% and had a market-model CAR of -9.9%. This is consistent with Germany's greater exposure to European energy and geopolitical risk. XLF also underperformed with a market-model CAR of -6.6%, while VGK and XLK were negative but less severe. JETS had a negative raw return but a small positive market-model CAR, meaning its poor raw performance was partly explained by overall market conditions and its normal market sensitivity.

The contrast between raw return and market-model CAR is important. For example, JETS looked weak in raw return terms, but once its high beta and broad market weakness are considered, its market-model CAR becomes slightly positive. This shows why the market model is useful: it prevents the analysis from confusing general market decline with sector-specific underperformance.

Table 3: Main $[-30, +30]$ Window Results

Rank	Ticker	Cum. Return	SPY Return	Simple CAR	Market CAR
1	XLE	+24.5%	-3.4%	+27.9%	+10.7%
2	ITA	+5.5%	-3.4%	+8.8%	+9.6%
3	XLB	-0.1%	-3.4%	+3.2%	+2.2%
4	JETS	-6.8%	-3.4%	-3.4%	+1.9%
5	XLK	-6.9%	-3.4%	-3.6%	-2.7%
6	VGK	-8.3%	-3.4%	-4.9%	-3.3%
7	XLF	-8.6%	-3.4%	-5.2%	-6.6%
8	EWG	-17.3%	-3.4%	-13.9%	-9.9%

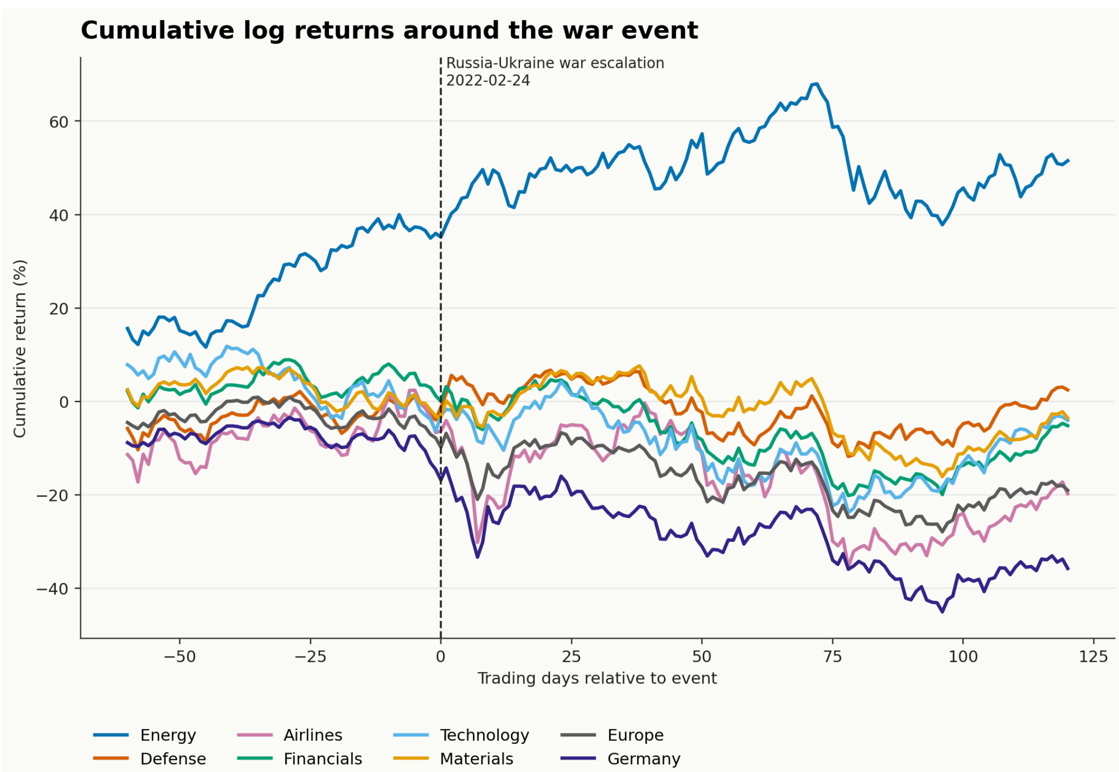

Figure 1: Cumulative returns around the Russia–Ukraine War event date.

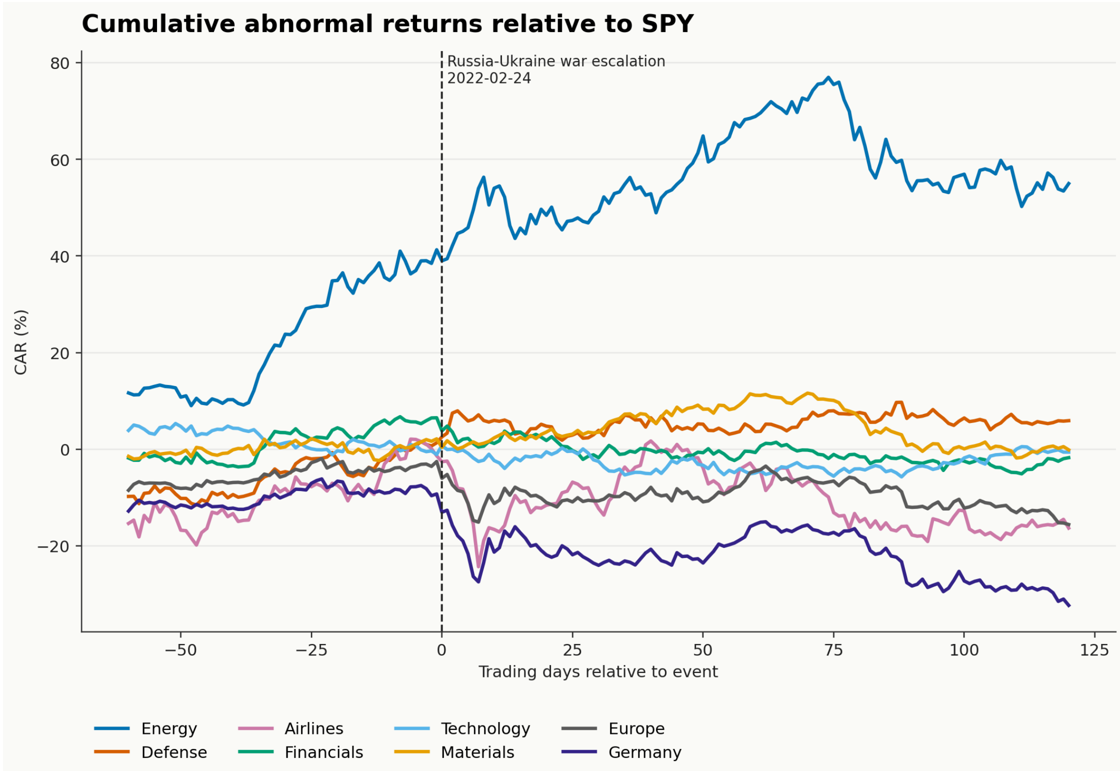

Figure 2: Cumulative abnormal returns relative to SPY.

3.4 Ranking, Window Comparison, and Volatility

The ranking chart reinforces the table results. XLE and ITA are the only large positive outliers in the main event window. EWG is the largest negative outlier, followed by XLF and VGK. This supports the idea that the war shock did not simply push the entire market in one direction. Instead, it redistributed market expectations across industries.

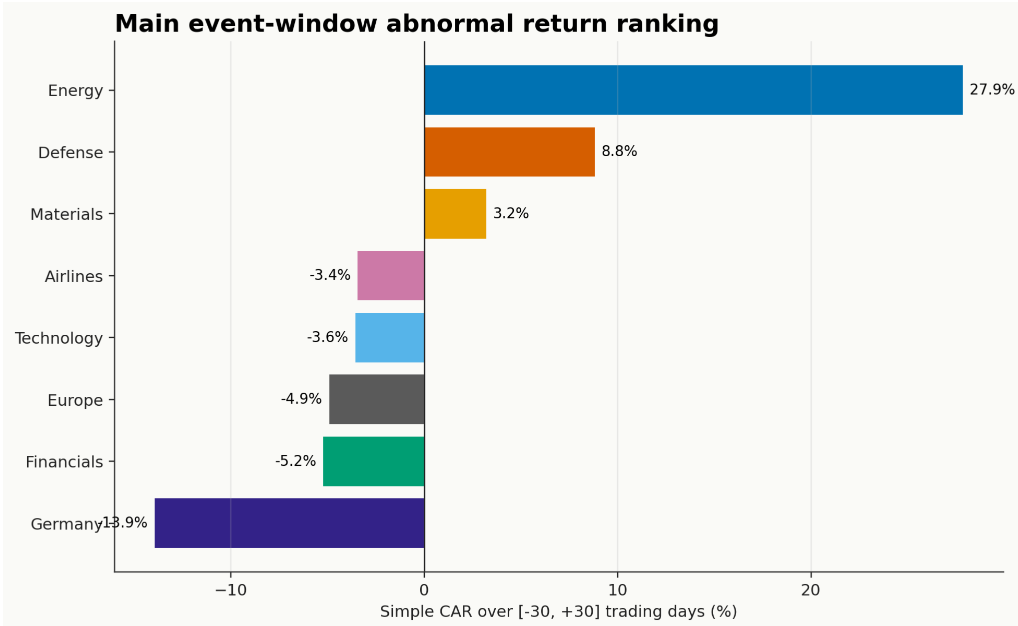

Figure 3: Main-window simple CAR ranking by ETF.

The three event windows show that the direction of results is broadly stable. XLE remains positive in the short, main, and post-event windows. ITA is especially strong in the short and main windows but smaller in the post-event window. This suggests that defense-related repricing may have been concentrated close to the invasion date. EWG remains negative across windows, which supports the interpretation that Germany had greater regional exposure to the shock.

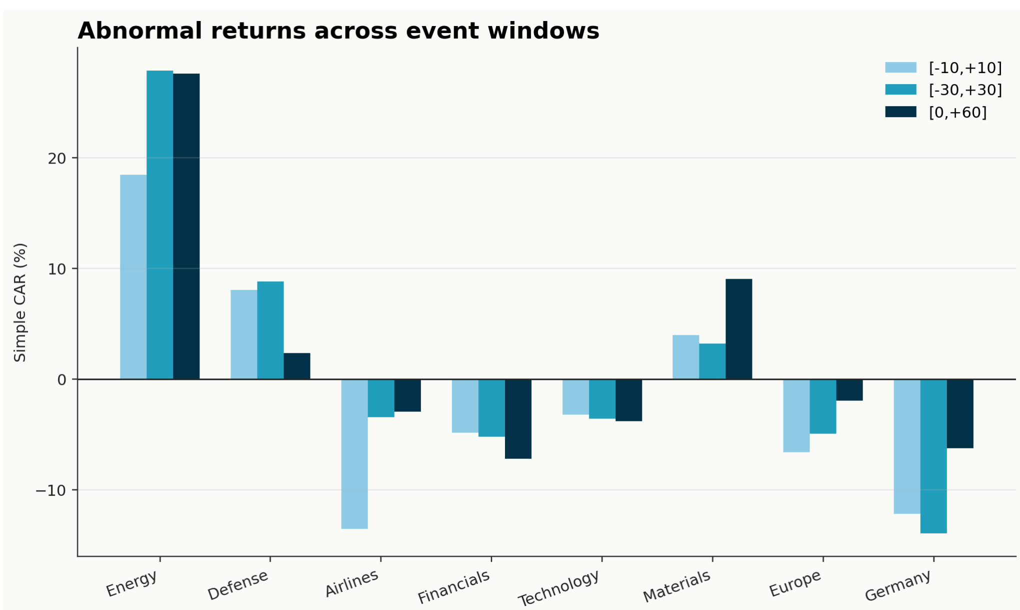

Figure 4: Comparison of simple CAR across event windows.

Volatility also matters. A sector can produce a large return but also carry large risk. JETS shows the highest volatility, which fits the uncertainty around air travel, fuel prices, and European airspace disruption. ITA has relatively low volatility compared with its positive CAR, making its event-window performance more stable than several other ETFs in the sample.

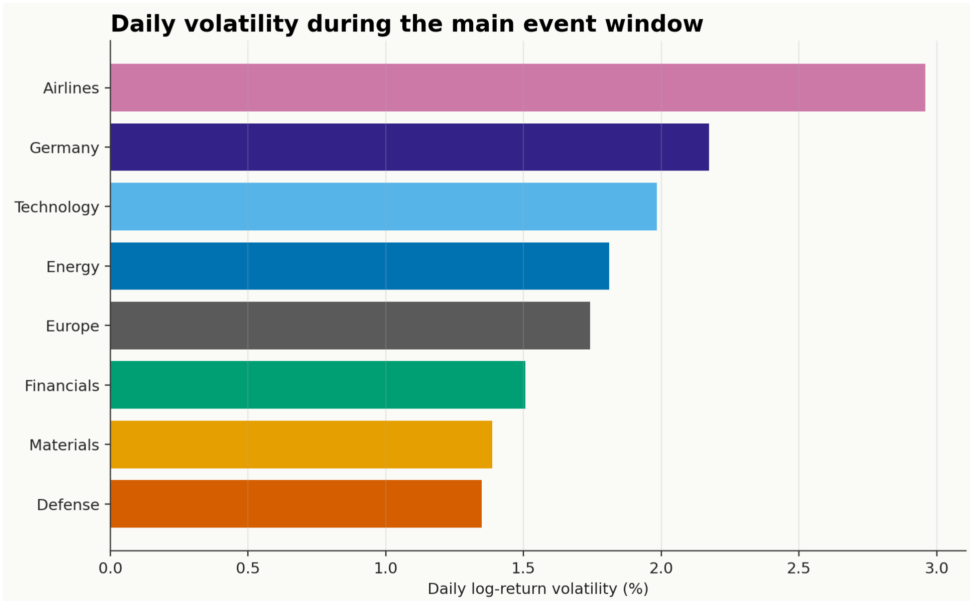

Figure 5: Main-window daily return volatility by ETF.

3.5 Difference-in-Differences Results

Table 4 reports the DiD regression. The interaction term, Post × Treatment, is +0.041% per day, but the t-statistic is only 0.10 and the 95% confidence interval ranges from -0.75% to +0.83%. Since the interval includes zero, the DiD result is not statistically significant.

Table 4: Difference-in-Differences Regression Results

Term	**Estimate**	**Std. Error**	**t-Statistic**	**95% CI**
Intercept	-0.00248	0.00143	-1.73	[-0.00529, +0.00032]
Post	+0.00230	0.00201	+1.15	[-0.00163, +0.00624]
Treatment	+0.00356	0.00286	+1.24	[-0.00205, +0.00917]
Post × Treatment	+0.00041	0.00401	+0.10	[-0.00746, +0.00827]

This does not mean the event study is wrong. Instead, it shows that the DiD model is less sensitive to a sudden shock spread across a noisy 60-day window. The event study captures abnormal returns close to the event date, while DiD averages returns before and after the event. If the market reaction was fast and front-loaded, the DiD estimate can become small even when the event-study CAR is visually and economically meaningful.

The DiD result also helps keep the conclusion balanced. The positive event-study CARs for XLE and ITA are large enough to discuss as meaningful market patterns, but the insignificant DiD coefficient means the report should avoid saying that the model proves a clean treatment effect. A better interpretation is that the event study detects sector-specific repricing, while the DiD model does not provide strong enough statistical confirmation over the broader window.

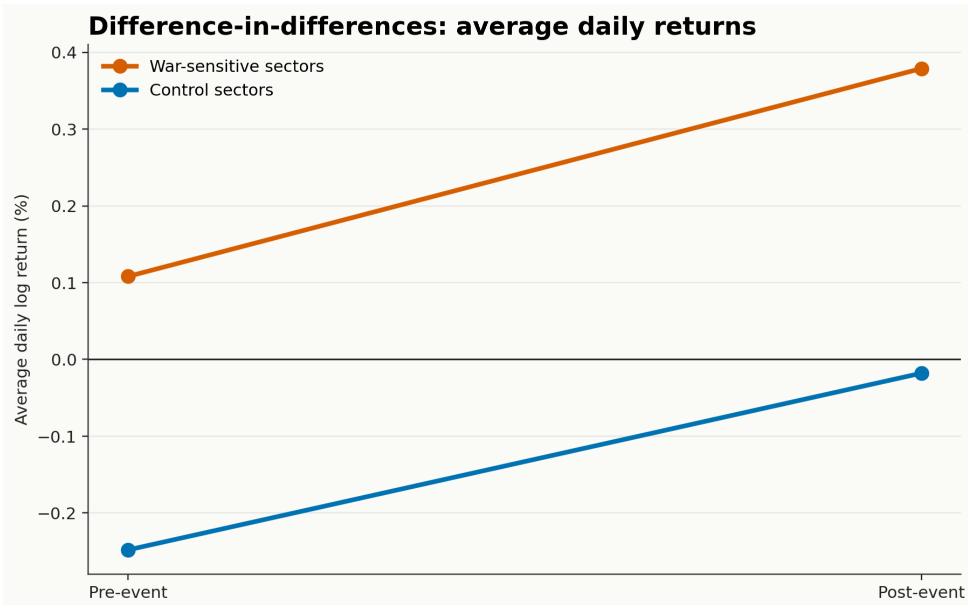

Figure 6: Mean daily returns for treatment and control groups before and after the event.

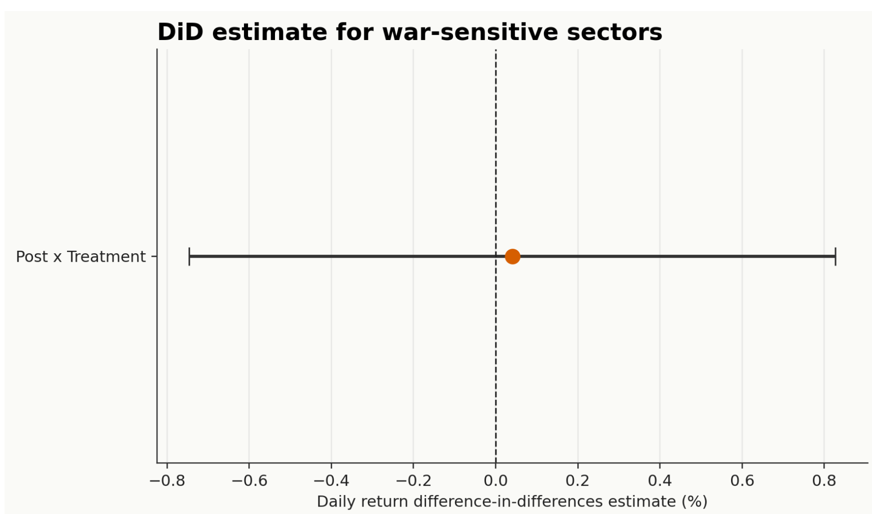

Figure 7: DiD interaction coefficient with 95% confidence interval.

3.6 Inflation and Market Context

The inflation chart is important because it reminds us that the war was not the only force affecting markets in 2022. U.S. and Euro Area inflation were already rising before the invasion, partly due to post-COVID demand recovery, supply-chain disruption, and commodity price pressure. The war intensified energy and food-price concerns, but the broader inflation environment means the results should not be interpreted as pure war causality.

This context is especially relevant for Energy. XLE's strong performance is consistent with the war shock, but energy prices were already elevated before the invasion. Similarly, airline performance was affected not only by war-related airspace and fuel concerns, but also by post-pandemic travel recovery. These overlapping conditions are why the report describes abnormal-return evidence rather than claiming the war was the only driver of sector performance.

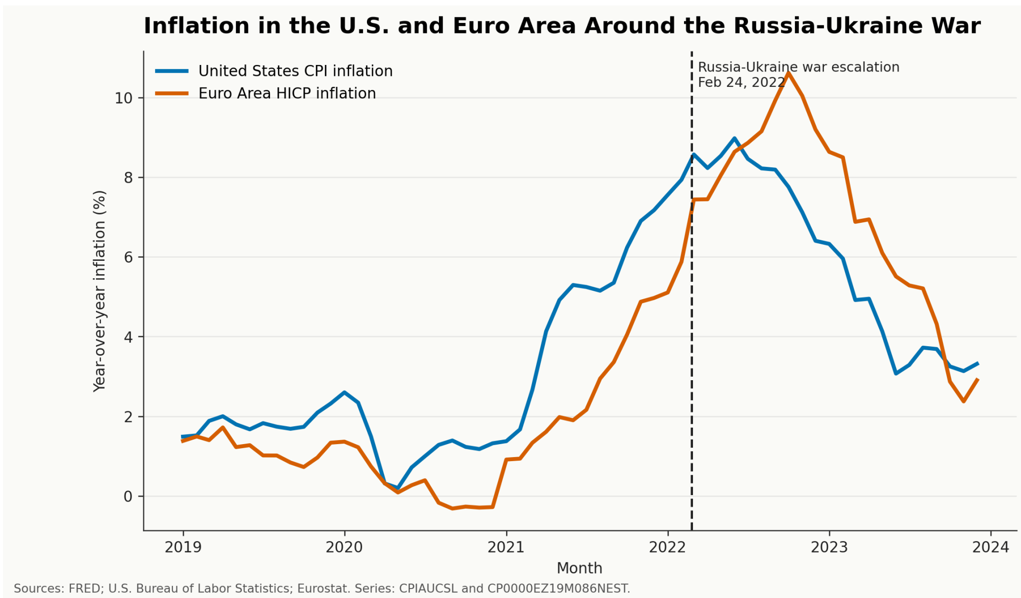

Figure 8: Inflation in the U.S. and Euro Area around the Russia–Ukraine War.

4 Discussion and Conclusion

The analysis leads to four main conclusions.

1. **Energy was the clearest outperformer.** XLE had the highest market-model CAR in the main event window, consistent with concerns over energy supply disruption and rising oil and gas prices.

2. **Defense also outperformed.** ITA showed a strong positive CAR, consistent with expectations of higher defense spending and increased demand for aerospace and defense firms.

3. **European exposure mattered.** EWG and VGK underperformed, with Germany showing the largest negative market-model CAR. This matches the idea that European markets faced more direct geographic and energy-related exposure.

4. **The statistical evidence is mixed.** The event study shows strong abnormal-return patterns, but the DiD interaction is not statistically significant. This means the project can identify meaningful market patterns, but it should avoid claiming definitive causal proof.

Overall, the event study suggests that investors reallocated exposure across sectors quickly around the invasion. The strongest positive abnormal returns were concentrated in Energy and

Aerospace and Defense, while the weakest results appeared in Germany, Europe, and Financials. The DiD model provides a useful caution: when a geopolitical event is sudden and noisy, a broad before-and-after comparison may fail to detect the same effect that an event-study window reveals.

The project has several limitations. First, SPY is a useful benchmark, but it does not control for all macroeconomic factors. Inflation, interest rates, oil prices, and post-pandemic recovery all overlapped with the invasion. Second, ETF-level data is broad and cannot show firm-level differences. Third, the event date is clear, but the conflict and related news developed over time, so a single event date simplifies a complex geopolitical process.

Another limitation is sample size. The analysis studies eight non-benchmark ETFs, which is enough for a focused class project but not enough to represent every industry or every country affected by the war. The DiD treatment group is also small because it contains only Energy and Aerospace and Defense. A larger treatment group could improve statistical power, but it might also make the treatment definition less precise.

Despite these limitations, the project provides a useful empirical framework. It shows how an investor or researcher can move from a broad news event to a structured statistical question: choose a benchmark, define exposed sectors, estimate expected returns, compute abnormal returns, and compare results across event windows. This structure makes the conclusions more disciplined than a simple visual comparison of price charts.

Future work could improve the analysis by adding a multi-factor model such as Fama-French, including oil and wheat futures, adding exchange rates and sovereign bonds, or estimating a time-varying DiD model. These extensions would help separate the Russia–Ukraine War shock from other macroeconomic forces and would provide a fuller view of how capital moved across markets during the crisis.

References

Federal Reserve Bank of St. Louis. FRED Economic Data: CPIAUCSL and Euro Area HICP inflation series. Data from the U.S. Bureau of Labor Statistics and Eurostat. Accessed June 2026.

Yahoo Finance chart data for SPY, XLE, ITA, JETS, XLF, XLK, XLB, VGK, and EWG. Downloaded with project Python scripts.